

Induction therapy with autologous mesenchymal stem cells in living-related kidney transplants: a randomized controlled trial.

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CONTEXT Antibody-based induction therapy plus calcineurin inhibitors (CNIs) reduce acute rejection rates in kidney recipients; however, opportunistic infections and toxic CNI effects remain challenging. Reportedly, mesenchymal stem cells (MSCs) have successfully treated graft-vs-host disease. **OBJECTIVE** To assess autologous MSCs as replacement of antibody induction for patients with end-stage renal disease who undergo ABO-compatible, cross-match-negative kidney transplants from a living-related donor. **DESIGN, SETTING, AND PATIENTS** One hundred fifty-nine patients were enrolled in this single-site, prospective, open-label, randomized study from February 2008-May 2009, when recruitment was completed. **INTERVENTION** Patients were inoculated with marrow-derived autologous MSC ($1-2 \times 10^6/\text{kg}$) at kidney reperfusion and two weeks later. Fifty-three patients received standard-dose and 52 patients received low-dose CNIs (80% of standard); 51 patients in the control group received anti-IL-2 receptor antibody plus standard-dose CNIs. **MAIN OUTCOME MEASURES** The primary measure was 1-year incidence of acute rejection and renal function (estimated glomerular filtration rate [eGFR]); the secondary measure was patient and graft survival and incidence of adverse events. **RESULTS** Patient and graft survival at 13 to 30 months was similar in all groups. After 6 months, 4 of 53 patients (7.5%) in the autologous MSC plus standard-dose CNI group (95% CI, 0.4%-14.7%; $P = .04$) and 4 of 52 patients (7.7%) in the low-dose group (95% CI, 0.5%-14.9%; $P = .046$) compared with 11 of 51 controls (21.6%; 95% CI, 10.5%-32.6%) had biopsy-confirmed acute rejection. None of the patients in either autologous MSC group had glucocorticoid-resistant rejection, whereas 4 patients (7.8%) in the control group did (95% CI, 0.6%-15.1%; overall $P = .02$). Renal function recovered faster among both MSC groups showing increased eGFR levels during the first month after surgery than the control group. Patients receiving standard-dose CNI had a mean difference of 6.2 mL/min per 1.73 m^2 (95% CI, 0.4-11.9; $P = .04$) and those in the low-dose CNI of 10.0 mL/min per 1.73 m^2 (95% CI, 3.8-16.2; $P = .002$). Also, during the 1-year follow-up, combined analysis of MSC-treated groups revealed significantly decreased risk of opportunistic infections than the control group (hazard ratio, 0.42; 95% CI, 0.20-0.85, $P = .02$). **CONCLUSION** Among patients undergoing renal transplant, the use of autologous MSCs compared with anti-IL-2 receptor antibody induction therapy resulted in lower incidence of acute rejection, decreased risk of opportunistic infection, and better estimated renal function at 1 year. **TRIAL REGISTRATION** clinicaltrials.gov Identifier: NCT00658073.